

PAKSE, Lao People's Democratic Republic

WMO: 489550

Lat: 15.136N

Lon: 105.779E

Elev: 335

StdP: 14.52

Time Zone: 7.00 (E07)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.4	63.9	50.1	54.4	70.9	52.2	58.8	71.9	21.2	76.0	17.5	73.1	4.4	0	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	15.9	98.1	78.0	96.3	77.6	94.8	77.2	81.4	90.8	80.8	89.9	80.3	89.1	4.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	153.0	86.0	78.5	150.0	85.0	78.1	147.9	84.6	45.4	91.5	44.8	90.5	44.3	89.5	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.4	12.8	11.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)				DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	77.9	81.5	85.5	86.9	85.4	83.6	82.0	81.8	82.0	81.8	80.4	78.2	(6)
(7)		DBStd	4.19	4.16	4.19	3.51	3.32	2.85	2.57	2.63	2.22	2.33	2.16	3.60	4.28	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11767	865	883	1099	1107	1098	1007	993	985	960	985	912	873	(10)
(11)		CDD65	6292	400	463	634	657	633	557	528	520	510	520	462	408	(11)
(12)		CDH74	69762	3929	5302	8172	8670	7976	6435	5445	5253	5149	5177	4410	3844	(12)
(13)		CDH80	29657	1665	2515	4298	4660	3827	2561	1775	1584	1628	1846	1721	1577	(13)
(14)	Wind	WSAvg	4.2	4.4	4.5	4.8	4.2	3.8	3.7	3.8	3.6	3.6	3.7	4.9	5.0	(14)
(15)	Precipitation	PrecAvg	87.3	0.3	0.6	1.8	3.3	9.5	10.7	19.5	19.2	15.7	5.4	1.2	0.4	(15)
(16)		PrecMax	111.0	1.9	2.8	5.0	8.2	18.4	19.4	34.4	30.6	37.0	12.7	4.3	3.9	(16)
(17)		PrecMin	58.2	0.0	0.0	0.4	1.1	4.5	3.9	6.4	10.0	8.9	1.1	0.1	0.0	(17)
(18)		PrecStd	14.0	0.5	0.7	1.2	1.9	3.4	4.3	6.8	5.7	6.4	2.9	1.0	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.9	96.8	98.8	101.9	99.1	95.1	93.5	91.8	92.6	93.5	94.0	94.4	(19)
(20)		MCWB	73.2	75.3	77.0	78.7	79.9	80.1	79.2	80.4	80.4	78.2	77.5	74.9	(20)	
(21)		2%	DB	92.1	94.8	97.0	99.1	96.1	92.6	91.1	89.8	90.5	91.3	92.1	92.4	(21)
(22)		MCWB	72.6	74.3	76.4	78.4	80.1	79.9	79.6	79.9	79.9	77.6	77.0	74.4	(22)	
(23)		5%	DB	90.4	93.0	95.4	96.8	94.2	91.0	89.2	88.2	89.0	90.0	90.3	90.3	(23)
(24)		MCWB	71.7	73.7	76.0	78.3	79.6	79.7	79.5	79.5	79.4	77.4	76.3	73.6	(24)	
(25)		10%	DB	88.3	91.2	94.0	94.8	92.3	89.5	87.4	86.7	87.5	88.3	88.5	88.0	(25)
(26)		MCWB	70.8	73.0	75.8	78.0	79.3	79.4	79.2	79.1	79.0	77.2	75.8	72.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	78.2	84.1	81.8	82.0	81.5	81.0	81.3	81.7	80.3	80.1	77.9	(27)
(28)		MCDWB	86.5	92.4	93.6	95.4	92.8	90.8	88.7	88.8	89.4	88.7	89.2	87.6	(28)	
(29)		2%	WB	75.5	76.5	79.3	80.4	81.0	80.8	80.3	80.7	80.7	79.5	79.1	76.3	(29)
(30)		MCDWB	85.6	90.7	91.8	93.4	91.8	89.6	87.7	87.7	88.2	87.4	88.5	87.2	(30)	
(31)		5%	WB	74.4	75.8	78.2	79.7	80.6	80.2	79.9	80.0	80.0	78.9	78.2	75.3	(31)
(32)		MCDWB	85.3	89.1	91.2	92.4	91.3	88.5	86.9	86.5	87.0	86.4	87.1	86.3	(32)	
(33)		10%	WB	73.2	74.8	77.3	79.0	80.0	79.8	79.3	79.5	79.5	78.2	77.3	74.3	(33)
(34)		MCDWB	84.8	87.4	90.0	91.5	90.0	87.7	85.9	85.4	86.0	85.2	85.1	85.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	21.5	20.4	18.0	15.9	13.6	11.0	10.0	9.8	10.5	13.4	16.1	19.5	(35)	
(36)		5% DB	MCDBR	22.4	20.8	18.8	17.5	15.9	12.8	12.7	12.2	12.9	15.9	17.6	20.4	(36)
(37)		MCWBR	6.8	6.5	6.0	4.9	5.1	4.3	4.6	4.6	4.9	5.7	6.0	6.5	(37)	
(38)		5% WB	MCDBR	18.9	17.5	16.4	15.9	14.2	11.6	11.3	11.1	11.9	13.4	15.9	18.9	(38)
(39)		MCWBR	6.9	6.4	6.4	5.6	5.1	4.5	4.8	4.8	5.0	5.2	6.1	7.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.439	0.564	0.664	0.621	0.498	0.464	0.460	0.460	0.483	0.483	0.427	0.415	(40)	
(41)		taud	2.250	1.900	1.707	1.852	2.209	2.316	2.322	2.313	2.243	2.223	2.367	2.354	(41)	
(42)		Ebn at Noon	270	240	221	231	257	263	265	268	262	259	272	274	(42)	
(43)		Edh at Noon	43	63	78	67	46	41	41	42	45	45	37	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1637	1742	1696	1734	1666	1497	1391	1347	1406	1550	1561	1538	(44)	
(45)		RadStd	73	76	60	79	82	102	143	107	149	132	79	70	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page